

Matrix-producing Carcinoma of the Breast

An Aggressive Subtype of Metaplastic Carcinoma

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Abstract: Matrix-producing carcinoma (MPC) of the breast is a subtype of metaplastic carcinoma defined as an invasive breast carcinoma with a direct transition of carcinoma to cartilaginous or osseous matrix without an intervening spindle cell component. Our aims were (1) to evaluate specific histologic characteristics of MPC and correlate these with disease recurrence; and (2) to determine whether rates of locoregional and distant recurrence for MPC are significantly different from those of invasive ductal carcinoma. Thirty-two cases of MPC were identified. Fourteen patients (44%) were ≤ 50 years of age; 10 (31%) had tumors of size ≤ 2 cm, and 6 (19%) had tumors ≥ 5 cm. In this series, all tumors contained chondromyxoid or chondroid matrix, and 1 (3.1%) also contained focal ($< 5\%$) osseous matrix. High-grade matrix was present in 9 cases (28%), and low-grade matrix was present in 23 (72%). Matrix comprised $\leq 10\%$ of the tumor in 14 cases (44%), $> 10\%$ but $< 40\%$ in 9 (28%), and $\geq 40\%$ in 9 (28%). The carcinomatous component was high grade in 30 cases (94%), and 19 tumors (59%) had central necrosis. Seven patients (22%) had positive axillary lymph nodes, and 8 (25%) had lymphovascular space invasion (LVSI). LVSI was the only factor independently associated with locoregional recurrence-free survival in multivariate analysis ($P = 0.043$). Although $\geq 40\%$ matrix was associated with improved distant recurrence-free (DFS) survival in univariate analysis ($P = 0.044$), only LVSI and tumor stage were independently associated with DFS survival in multivariate analysis ($P = 0.027$ and $P = 0.001$, respectively). Compared with matched controls with invasive ductal carcinoma, patients with MPC had decreased locoregional recurrence-free survival ($P = 0.001$) and decreased DRF survival ($P = 0.001$). In summary, MPC is an aggressive subtype of metaplastic carcinoma with a worse clinical outcome than invasive ductal carcinoma.

Key Words: matrix-producing carcinoma, metaplastic carcinoma, breast cancer, locoregional recurrence, distant recurrence

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Matrix-producing carcinoma (MPC) of the breast, first described by Wargotz and Norris in 1989,²³ is a subtype of metaplastic carcinoma defined as an invasive breast carcinoma with a direct transition of carcinoma to cartilaginous or osseous matrix without an intervening spindle cell component. MPC is just 1 entity that falls under the rubric of metaplastic carcinoma, a heterogeneous group of tumors that also includes low-grade fibromatosis-like spindle cell carcinoma,^{6,22} sarcomatoid carcinoma,^{1,2,5,27} metaplastic carcinoma with osteoclastic giant cells,²⁶ squamous cell carcinoma,^{4,17,19,25} adenosquamous carcinoma,^{3,20} mucoepidermoid carcinoma analogous to those occurring in the salivary gland^{8,9,11} and carcinosarcoma.^{19,24}

Given the heterogeneous composition of these tumors and the fact that metaplastic carcinomas are rare, comprising less than 1% of invasive breast carcinomas,¹⁰ outcome data are limited and may seem contradictory. However, some subtypes seem to have distinct biologic behavior. Low-grade fibromatosis-like spindle cell carcinoma has a relatively favorable outcome,^{6,22} while carcinosarcoma is a highly aggressive tumor.²⁴ In the original description of MPC by Wargotz and Norris,²³ the authors concluded that the outcome of patients with MPC is not significantly different from those with invasive ductal carcinomas of similar grade and stage. Our more recent anecdotal experience, however, suggested that these were more aggressive tumors.

In this study, we examined cases of MPC diagnosed at our institution and evaluated whether the local and distant recurrence rates for MPC were significantly different from those of invasive ductal carcinoma. We also evaluated whether specific histologic characteristics of MPC were associated with tumor recurrence.

MATERIALS AND METHODS

After receiving approval from the Institutional Review Board of M.D. Anderson Cancer Center (MDACC), cases of MPC of the breast diagnosed at MDACC from 1989 to 2007 were identified by searching the surgical pathology database using the SNOMED code for *matrix-producing carcinoma* and by performing a text search of breast pathology report diagnoses using the terms *metaplastic carcinoma* and *matrix*. All slides were retrieved from the MDACC pathology archives. Inclusion criteria included patients with: (1) a primary invasive carcinoma of the breast; (2) an invasive breast carcinoma with a

direct transition of carcinoma to cartilaginous or osseous matrix without an intervening spindle cell component; and (3) available clinical follow-up. Cases were excluded if representative material from the primary tumor was not available for review.

The morphologic features evaluated included the type of matrix, the percentage of matrix comprising viable tumor, the grade of the matrix component, the distribution of matrix, the percentage of carcinoma comprising viable tumor, the grade of the carcinomatous component, the shape of the tumor, the presence or absence of central or patchy necrosis, and the percentage of necrosis comprising the tumor. The matrix was classified as either chondroid or osseous and graded as low or high grade. Low-grade matrix cells had small bland nuclei with smooth nuclear contours, and high-grade matrix cells contained enlarged irregular nuclei with coarse chromatin and identifiable nucleoli. The distribution of matrix was classified as either diffuse (throughout the lesion in a continuous fashion), unifocal, or multifocal (scattered as discontinuous foci). The shape of each tumor was nodular, multinodular, or raggedly invasive. The modified Scarff-Bloom-Richardson grading scheme was employed to grade the carcinomatous component. Lymphovascular space invasion (LVSI) was recorded, and the presence of adjacent ductal carcinoma in situ was noted.

Clinical information was obtained from the pathology reports, the medical record and the MDACC Surgical Oncology database. Recurrence data for a 2:1 matched control group (2 control patients for every case) were also obtained from the same database. Control patients were matched for patient age, stage, and tumor grade. For each patient with MPC, the 2 matched control patients who had surgery performed closest to the date of surgery of the corresponding patient with MPC were selected. Locoregional recurrence-free (LRF) and distant recurrence-free (DRF) survival estimates were calculated using the Kaplan-Meier product limit method and were expressed $\pm$ standard error. The 2-sided log-rank test was used to test the association between particular factors and LRF and DRF survival. Multivariate analysis was performed using the Cox proportional hazards regression model. All statistical analyses were carried out using SPSS 12.0 for Windows (SPSS Inc, Chicago, IL). LRF survival was defined as the interval from the date of surgery to the date of first locoregional disease recurrence or to the last follow-up date. All locoregional recurrences were scored as events regardless of the presence of distant metastatic disease, and patients without recurrence were censored at last follow-up. DRF survival was defined as the interval from the date of surgery to the date of the first distant recurrence.

RESULTS

Thirty-two cases that fulfilled the inclusion criteria were available for study. All patients were female and ranged in age from 38 to 71 years (median age 50 y) at the time of diagnosis. Primary tumor size ranged from 1.0 to 8.0 cm (median tumor size 3.0 cm). The number of

hematoxylin and eosin-stained slides evaluated for each primary tumor ranged from 1 to 11 (median number 4). The pathologic features of the 32 cases are summarized in Table 1. All cases by definition consisted of invasive carcinoma with an abrupt transition to tumor matrix without an intervening spindle cell sarcomatoid component (Fig. 1A). In this series, all cases contained at least focal areas of chondromyxoid matrix, which comprised 2% to 98% of the viable tumor (median 18%).

TABLE 1. Pathologic Features of the 32 MPCs Included in the Study

Pathologic Features	No. Patients (%)
Tumor size	
≤ 2 cm	10 (31)
> 2 cm and ≤ 5 cm	16 (50)
> 5 cm	6 (19)
Axillary lymph node status	
Negative	25 (78)
Positive	7 (22)
Grade of matrix component	
Low	23 (72)
High	9 (28)
Grade of carcinomatous component	
Low	1 (3)
Intermediate	1 (3)
High	30 (94)
Matrix distribution	
Unifocal	3 (9)
Multifocal	16 (50)
Diffuse	13 (41)
Tumor shape	
Nodular	15 (50)
Multinodular	10 (33)
Raggedly infiltrating	5 (17)
Percentage of matrix	
$\leq 10\%$	14 (44)
$> 10\%$ and $< 40\%$	9 (28)
$\geq 40\%$	9 (28)
Percentage of carcinoma	
$\leq 10\%$	3 (9)
$> 10\%$ and $< 40\%$	16 (50)
$\geq 40\%$	13 (41)
Necrosis	
Absent	8 (25)
Patchy	5 (16)
Central	19 (59)
Lymphovascular invasion	
Absent	24 (75)
Present	8 (25)
Associated DCIS	
Absent	23 (72)
Present	9 (28)
ER	
Negative	26 (84)
Positive	5 (16)
PR	
Negative	29 (94)
Positive	2 (6)
HER2	
Negative	29 (100)
Positive	0 (0)

DCIS indicates ductal carcinoma in situ; ER, estrogen receptor; MPC, matrix-producing carcinoma; PR, progesterone receptor.

In addition, focal areas of the matrix ranged from mucoid appearing (Fig. 1B) to more overtly cartilaginous with hyaline matrix (Fig. 1C). Foci of hyaline cartilage were

noted in 6 cases (19%), consisting of either discrete nodules (Fig. 1D) or small microscopic foci in a background of chondromyxoid matrix (Fig. 1E). The chondromyxoid or

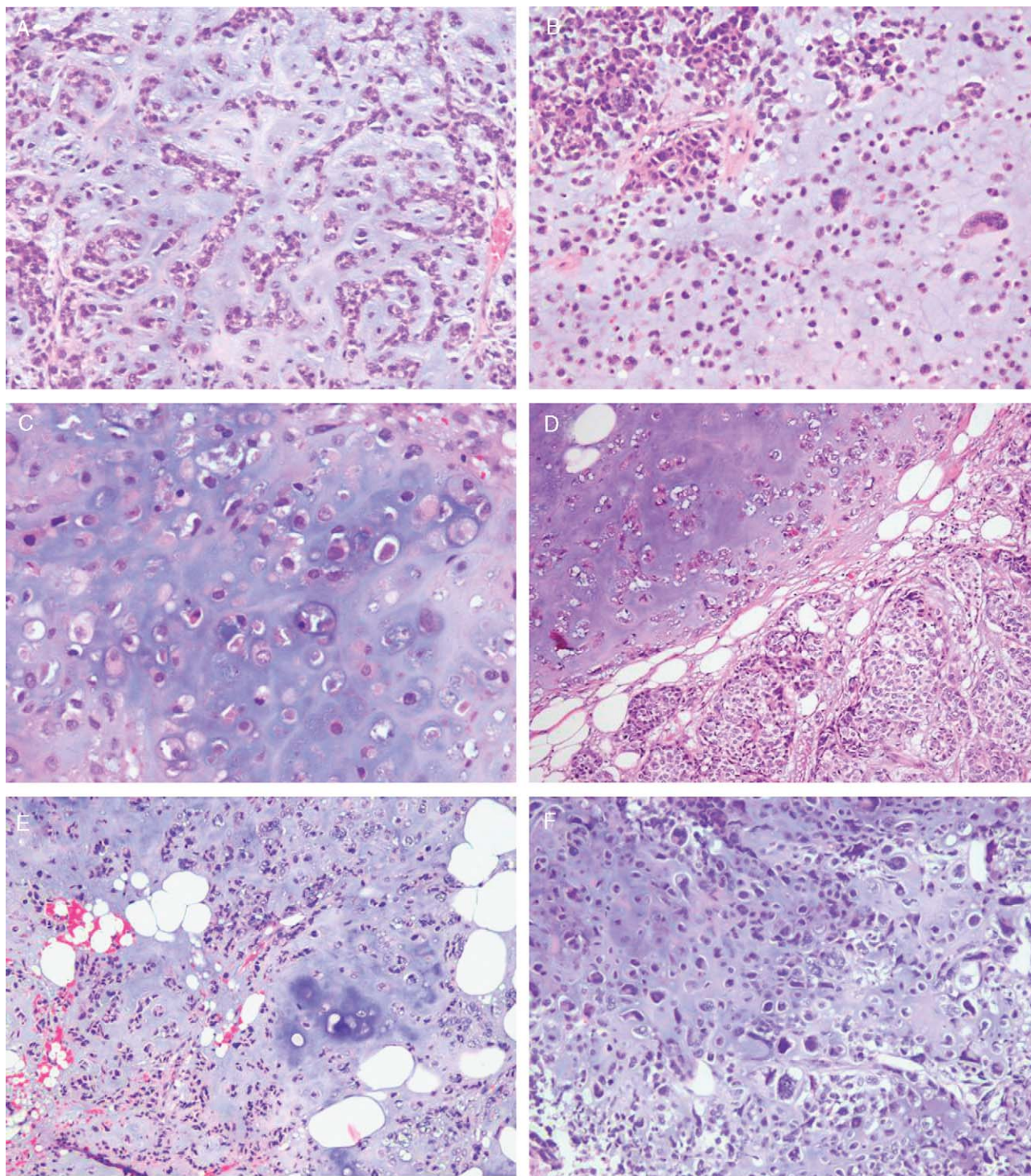


FIGURE 1. A, Transition from carcinoma to low-grade chondromyxoid matrix without intervening spindle cell component (case 7); B, Focal area of matrix with mucoid appearance (case 31); C, Hyaline cartilaginous matrix (case 9), producing (D) a prominent nodule (case 19) or (E) appearing as a small microscopic focus in a background of chondromyxoid matrix (case 19); F, High-grade chondroid matrix (case 27).

chondroid matrix was low grade in 23 cases (72%) (Fig. 1A) and high grade in 9 (28%) (Fig. 1F). Only 1 case in this series (3.1%) also contained a focal osseous component comprising < 5% of the matrix (Fig. 2A).

Matrix distribution was predominantly either multifocal (16 cases, 50%) or diffuse (13 cases, 41%), with 3 cases (9%) having unifocal matrix. Five tumors (16%) had patchy necrosis (Fig. 2B), and 19 cases (59%) had

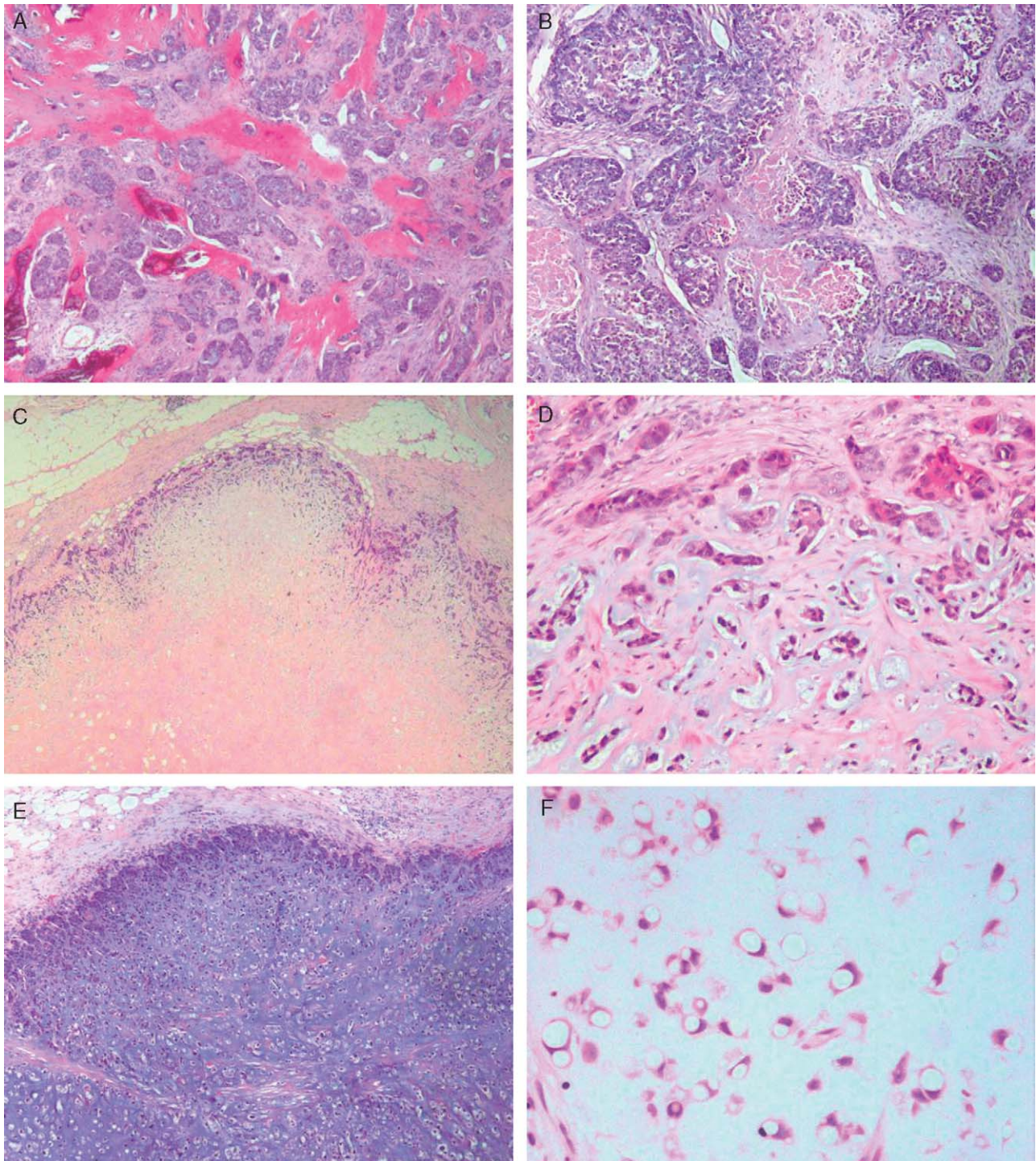


FIGURE 2. A, Focal osseous component (case 14); B, Patchy necrosis in carcinomatous component (case 14); C, Prominent central necrosis with peripheral rim of viable tumor showing (D) chondromyxoid bands enveloping small nests and clusters of viable carcinoma cells at the periphery (case 6); E, Large central area of chondromyxoid matrix and small rim of surrounding carcinoma cells (case 28); F, Signet ring cells embedded within chondroid nodule (case 3).

central necrosis comprising 1% to 75% of the tumor (median 24%). Six cases (19%) had prominent central necrosis comprising > 50% of the tumor, with a surrounding rim of viable neoplasm (Fig. 2C). The matrix in these tumors with prominent central necrosis generally consisted of chondromyxoid bands that enveloped small nests and clusters of viable carcinoma cells at the periphery (Fig. 2D).

One tumor contained nodules with large central areas of chondromyxoid matrix and small rims of surrounding carcinoma cells (Fig. 2E), producing an appearance very similar to those tumors with prominent central necrosis, but with central matrix instead. Another tumor had signet ring cells embedded within chondroid nodules (Fig. 2F).

In almost all cases, the carcinomatous component was high grade (30 cases, 94%). Tumor borders were evaluable in 30 cases. A minimally invasive border was present in 15 cases (50%), a multinodular pushing border was present in 10 (33%), and 5 (17%) had a raggedly infiltrative border. A ductal carcinoma in situ component

was present adjacent to or admixed with the primary tumor in 9 cases (28%). Eight cases (25%) had LVSI.

Hormone receptor status was available for 31 patients. Twenty-six (84%) were estrogen receptor negative, and 29 (94%) were progesterone receptor negative. All 29 patients with known HER2 status were negative for HER2 overexpression. Seven of the 32 patients (22%) had axillary lymph node metastases, with metastases from 4 of the 6 patients available for review composed purely of the carcinomatous component. A lymph node metastasis from 1 patient had a histologic appearance similar to that of the primary tumor, with 40% of the metastasis composed of chondromyxoid matrix. The lymph node metastasis from the remaining patient was predominantly carcinomatous with a 1-mm focus (< 5%) of chondromyxoid matrix.

Clinical features of the patients in this series are presented in Table 2. Fourteen patients underwent mastectomy and 18 underwent breast-conserving surgery (BCS). All patients were considered to have negative margins (2 mm or greater). Sixteen patients who underwent

TABLE 2. Clinical Features of the 32 Patients With MPC Included in the Study

Case No.	Patient Age	Surgical Procedure	Axillary Nodal Status*	Radiation Therapy†	Chemotherapy‡	Follow-up (mo)
1	58	Mastectomy	0/18	No	No	NED (98)
2	50	BCS	0/29	Yes	Adjuvant	NED (72)
3	69	Mastectomy	1/10	Yes	Neoadjuvant and adjuvant	NED (90)
4	44	Mastectomy	3/14	Yes	Neoadjuvant and adjuvant	LRR (10); positive pleural fluid (10); DOD (13)
5	48	BCS	2/2	After LRR	Adjuvant	LRR (24); NED (46)
6	47	BCS	0/12	Yes	Adjuvant	Lung metastasis (25); DOD (29)
7	50	BCS	0/4	Yes	Adjuvant	NED (45)
8	38	Mastectomy	4/14	Yes	Neoadjuvant and adjuvant	Lung metastasis (29); DOD (43)
9	40	BCS	0/9	Yes	Adjuvant	LRR (22); DOD (50)
10	52	Mastectomy	0/6	No	Neoadjuvant and adjuvant	Lung, mediastinum, bone metastases (29); DOD (29)
11	49	Mastectomy	8/22	Yes	Neoadjuvant and adjuvant	Positive pleural fluid, bone and thyroid metastases (11); DOD (11)
12	38	BCS	1/19	Yes	Neoadjuvant and adjuvant	NED (34)
13	57	Mastectomy	0/2	No	Adjuvant	NED (41)
14	42	BCS	0/6	Yes	Adjuvant	NED (26)
15	59	BCS	0/2	Yes	Adjuvant	NED (16)
16	63	BCS	0/2	Yes	Adjuvant	LRR (17); lung metastasis (17); AWD (17)
17	43	BCS	23/31	Yes	Neoadjuvant and adjuvant	NED (12)
18	50	BCS	0/1	Yes	Neoadjuvant	NED (10)
19	43	BCS	0/2	Yes	Adjuvant	NED (8)
20	62	BCS	0/1	Yes	Adjuvant	NED (3)
21	59	BCS	0/3	Yes	Neoadjuvant	NED (10)
22	51	Mastectomy	0/6	No	Adjuvant	NED (48)
23	38	Mastectomy	0/38	No	Adjuvant	LRR (56); NED (94)
24	49	Mastectomy	0/multiple	No	Neoadjuvant and adjuvant	LRR (1); Lung metastasis (1); DOD (19)
25	64	Mastectomy	0/9	No	Adjuvant	NED (7)
26	38	Mastectomy	4/18	Yes	Adjuvant	NED (34)
27	42	Mastectomy	0/6	No	Adjuvant (only initiated trx after distant metastasis)	Soft tissue of buttocks metastasis (29); AWD (29)
28	64	BCS	0/3	Yes	Adjuvant	NED (36)
29	71	BCS	0/3	Yes	None	NED (9)
30	54	BCS	0/3	After LRR	Adjuvant	LRR (26); lung metastasis (40); AWD (40)
31	38	BCS	0/19	Yes	Neoadjuvant	Positive pleural fluid (13); DOD (15)
32	53	Mastectomy	0/19	Yes	Neoadjuvant and adjuvant	NED (44)

*Number of axillary lymph node metastases/total number of axillary lymph nodes examined.

†Radiation therapy given after primary surgery and chemotherapy unless otherwise specified.

‡Chemotherapy regimen for breast carcinoma consisted of various combinations of 5-fluorouracil, doxorubicin, cyclophosphamide, docetaxel, paclitaxel, and carboplatin.

AWD indicates alive with disease; BCS, breast-conserving surgery; DOD, died of disease; LRR, locoregional recurrence; MPC, matrix-producing carcinoma; NED, no evidence of disease.

BCS received postoperative radiation therapy at the time of their initial treatment. Six patients who had mastectomy received radiation therapy. Twelve patients received neoadjuvant chemotherapy, 9 of whom also received adjuvant chemotherapy. Eighteen received adjuvant chemotherapy alone. Follow-up ranged from 3 to 98 months (median 29 mo).

At last follow-up, 8 patients had died of disease, with time to death ranging from 11 to 50 months (median 29 mo). Seven patients developed locoregional recurrence 1 to 56 months after surgery (median 25 mo), and 10 patients developed distant metastasis 1 to 40 months after surgery (median 28 mo). Three of the patients who died of disease had developed both locoregional and distant recurrence. Six of the 10 patients who developed distant recurrence had received total mastectomy, whereas 4 had received BCS. Four of the 7 patients who developed locoregional recurrence did not receive radiation therapy at the time of their initial treatment, and 2 of these 4 patients had received BCS (the only 2 patients who received BCS without radiation).

A lung metastasis and/or positive pleural fluid, observed in 6 patients, was the most common manifestation of distant disease. The remaining sites of distant

metastasis were the liver in 2 patients, the soft tissue of the buttocks in 1 patient, and the leptomeninges in 1 patient. Two of the patients with metastases had widespread disease. The diagnosis of metastatic disease was based on clinicoradiologic findings in 6 cases, whereas 4 cases had pathology available for review, including 2 pleural fluids, 1 lung metastasis, and 1 metastasis to the soft tissue of the buttocks. The lung metastasis available for review had chondromyxoid bands surrounding nests and individual tumor cells. This lung metastasis was from a patient whose primary tumor had prominent central necrosis with only minimal chondromyxoid matrix deposition at the periphery (Fig. 2D). The metastasis to the soft tissue of the buttocks also contained both a carcinomatous and chondromyxoid matrix component.

Clinicopathologic features associated with decreased LRF survival for patients with MPC were LVSI and stage IV disease ($P = 0.003$ and $P < 0.001$, respectively). LVSI, stage IV disease, and patient age ≤ 50 were each associated with decreased DRF survival ($P = 0.012$, $P < 0.001$, and $P = 0.033$, respectively), whereas matrix comprising $\geq 40\%$ of viable tumor was associated with increased DRF survival ($P = 0.044$) (Figs. 3A–D). The favorable outcome of patients with matrix comprising

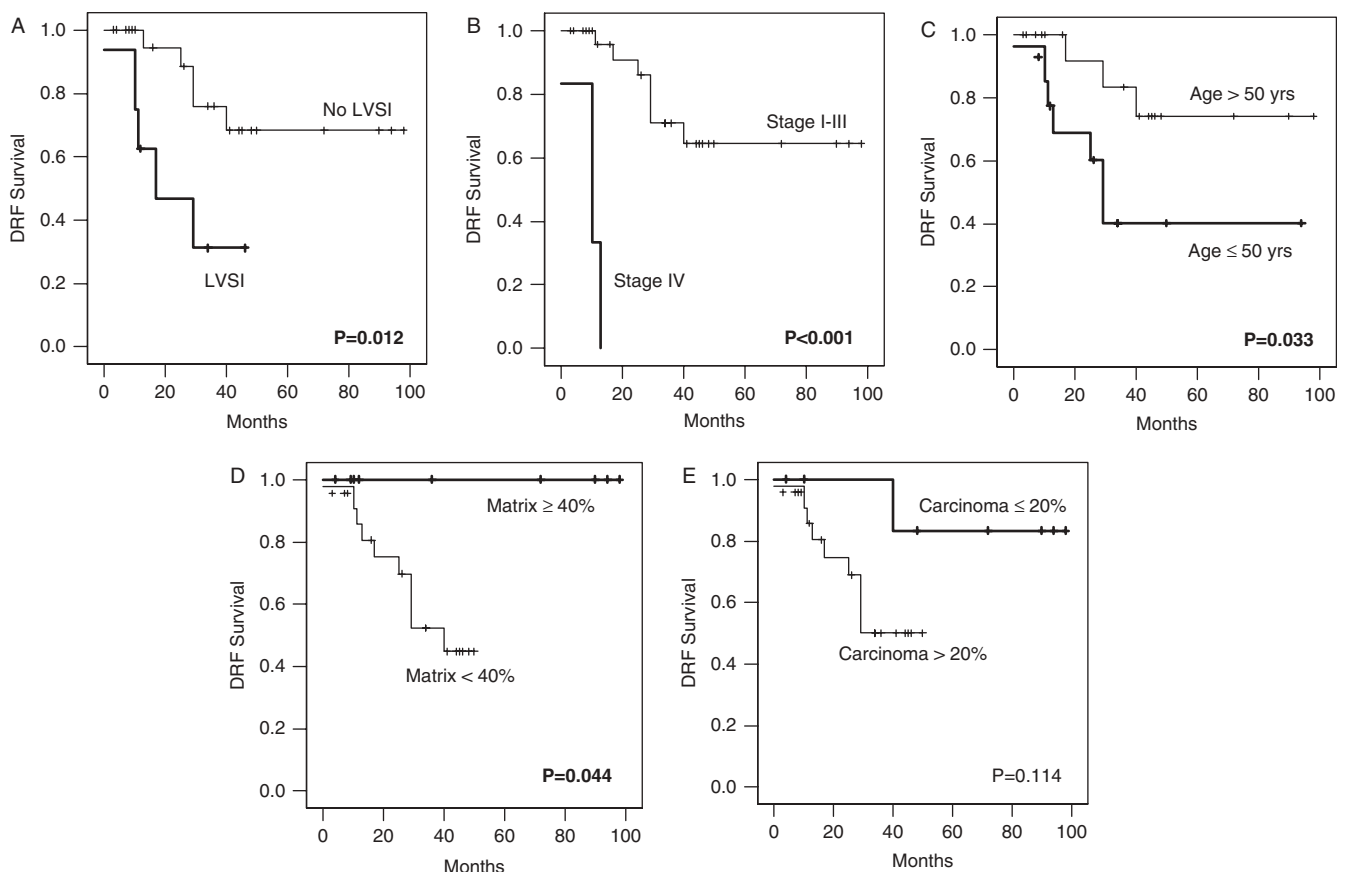


FIGURE 3. Distant recurrence-free (DRF) survival for patients with MPC according to: A, LVSI, B, Tumor stage, C, Patient age, D, Percent of viable tumor composed of matrix, and E, percent of viable tumor composed of carcinoma cells. LVSI indicates lymphovascular space invasion.

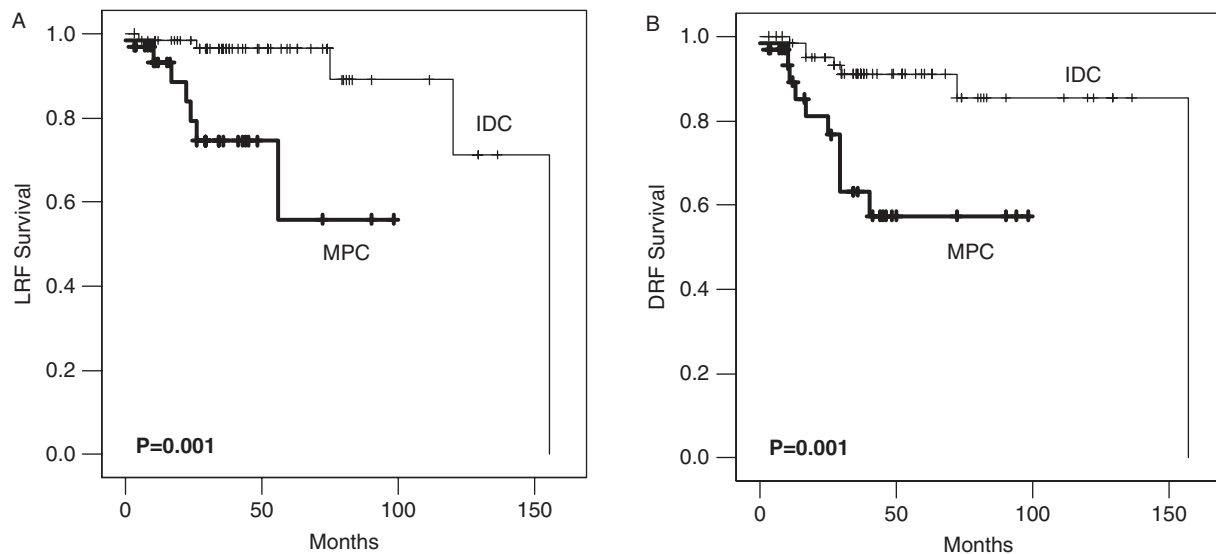


FIGURE 4. A, Locoregional recurrence-free (LRF) and B, Distant recurrence-free (DRF) survival for patients with MPC versus invasive ductal carcinoma (IDC). MPC indicates matrix-producing carcinoma.

≥40% of viable tumor might be explained in part by less tumor cellularity, although the association between DRF survival and the percentage of tumor composed of carcinoma cells did not reach statistical significance (Fig. 3E). In multivariate analysis, LVSI was the only factor independently associated with LRF survival [hazard ratio (HR) 6.54; 95% confidence interval (CI), 1.06-40.4; $P = 0.043$], whereas factors independently associated with DRF survival were LVSI (HR 4.76; 95% CI, 1.19-19.0; $P = 0.027$) and stage IV disease (HR 60.1; 95% CI, 5.07-712; $P = 0.001$).

Compared with a 2:1 matched control group of patients with invasive ductal carcinoma (2 control patients for each case; controls matched for patient age, stage, and tumor grade), patients with MPC had decreased LRF survival ($P = 0.001$) and decreased DRF survival ($P = 0.001$) (Fig. 4). The 9-year actuarial LRF and DRF survival for patients with MPC compared with control patients was $57\% \pm 16\%$ vs. $88\% \pm 8\%$ and $56\% \pm 11\%$ vs. $84\% \pm 8\%$, respectively.

DISCUSSION

Metaplastic carcinoma of the breast is a term that encompasses numerous distinct histologic entities with heterogeneous morphology. Given the rarity of metaplastic breast carcinoma, many studies to date have not subcategorized these tumors into specific entities, but rather, have looked at them as a collective group. In a recent study using data from the National Cancer Database, for example, the clinical presentation and patterns of treatment of metaplastic breast carcinoma ($n = 892$) were compared with matched cases of invasive ductal carcinoma ($n = 255,164$). The authors concluded that metaplastic breast carcinoma is generally hormone receptor negative and presents with larger tumor size, less nodal involvement,

and higher tumor grade compared with invasive ductal carcinoma. They also reported that, although metaplastic breast carcinoma is treated using the same principles as invasive ductal carcinoma, metaplastic carcinoma is generally treated more aggressively (with mastectomy and chemotherapy) because of a higher stage at presentation.¹⁸

Although many metaplastic carcinomas share a number of clinicopathologic characteristics, there are data indicating that some subtypes have distinct biologic behavior. For example, low-grade fibromatosis-like spindle cell carcinoma, a subtype of metaplastic breast carcinoma, has a relatively indolent behavior. Gobbi et al⁶ observed a tendency for local recurrence, but none of the tumors in their study metastasized. Sneige et al²² similarly reported a tendency for local recurrence and an absence of axillary lymph node metastases. Although 2 of the patients in their study did develop lung metastases within 2 years of diagnosis, this represents only 3.7% of the 54 patients in the 2 studies combined.

In contrast, Davis et al² examined the clinicopathologic features of patients diagnosed with metaplastic carcinoma of the breast composed of pure or almost pure sarcomatoid morphology and found a high rate of both local recurrence and distant metastasis. Ten of the 21 patients (48%) developed local relapse, and 9 of 21 (43%) developed distant metastases, with 66% of those having previously experienced a local failure. The median interval between diagnosis and local recurrence was 8.5 months (range, 4 to 76 mo), and the mean time to development of distant metastasis was 18 months (range, 3 to 90 mo). The most common site of distant metastasis was the lung. A similarly aggressive course for the subtype of metaplastic carcinoma composed predominantly of sarcomatoid carcinoma was observed by Carter et al.¹

There are very limited data available on the clinical behavior of MPC, a more unusual but distinct morphologic

subtype of metaplastic carcinoma of the breast. In the original study of MPC by Wargotz and Norris,²³ 26 patients were included with a minimum follow-up of 5 years. The patients ranged in age from 35 to 84 years (mean 58 y), and tumor size ranged from 0.5 to 12 cm (mean 3.9 cm). The clinical outcome was reported to be similar to that of a matched control group of patients with invasive ductal carcinoma. Features of MPC associated with worse outcome were diffuse cellularity of the matrix and atypical cartilaginous matrix.

Since the original study by Wargotz and Norris,²³ there have been only very small series or case reports of MPC.^{7,12–16,21} Our study represents the largest series to date. In contrast to the original study, we found MPCs to be significantly more aggressive than invasive ductal carcinomas matched for patient age, stage, and tumor grade. Both locoregional and distant recurrence rates for MPC were strikingly high, especially considering that most patients in our study were staged as T2N0. In addition to having somewhat smaller tumors, the patients in our study were also younger than those reported by Wargotz and Norris.²³ Moreover, the cases reported in the original study included several with osteoid in addition to cartilaginous matrix. Overt carcinoma was only a minor component of the tumors in the study by Wargotz and Norris,²³ whereas the carcinomatous component was the major component in most of our cases. Tumors with matrix comprising $\geq 40\%$ of the viable tumor in our study had improved DRF survival by univariate analysis.

Wargotz and Norris²³ reported that atypical cartilaginous matrix was an adverse prognostic factor among patients with MPC. Atypical cartilaginous matrix seems to correspond to what we classified as high-grade matrix, but the grade of the matrix in our study did not correlate with tumor recurrence. Tumors with a high proportion of matrix ($\geq 40\%$) had a more favorable clinical course in our study. We thought that this might simply reflect a smaller number of carcinoma cells per tumor volume, and although we did observe decreased recurrences for tumors with a small proportion of carcinoma cells, the latter did not reach statistical significance. Neither the proportion of matrix nor tumor cellularity had independent prognostic significance in multivariate analysis.

In summary, MPC is an aggressive variant of metaplastic breast carcinoma with increased locoregional and distant tumor recurrence compared with invasive ductal carcinoma. Postoperative radiation therapy after BCS is probably important for controlling local disease, as the only 2 patients who underwent BCS without postoperative radiation in our study developed locoregional recurrence. Aggressive systemic therapy seems to be warranted, as there is a high rate of distant recurrence despite the frequent absence of lymph node metastasis at presentation.

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